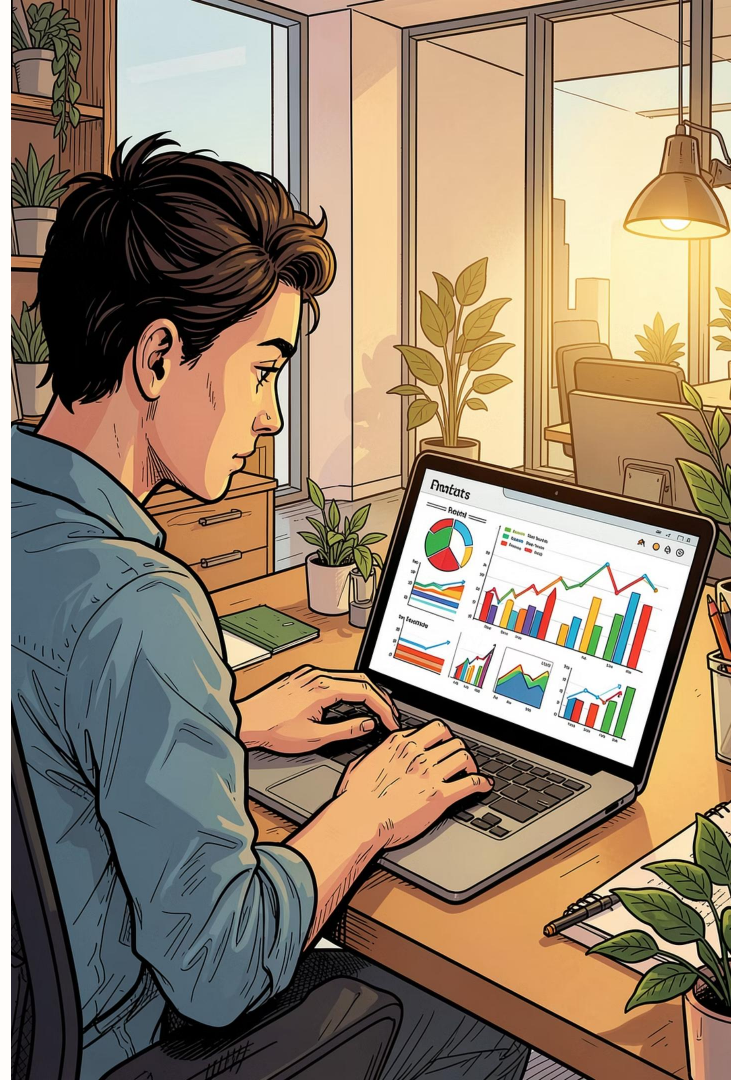


Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and preferences from **3,900 transactions** to guide strategic decisions.



Dataset at a Glance

3,900

Transactions

18

Columns

37

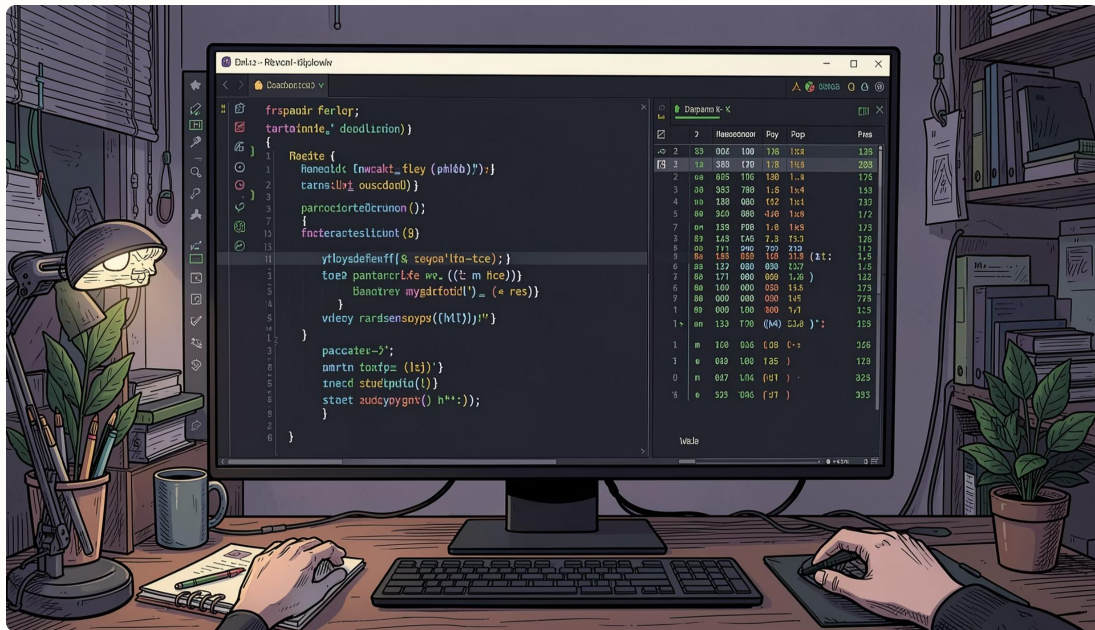
Missing Values

Review Rating column

Features span demographics, purchase details, and shopping behavior.



Python: Data Preparation



→ Load & Explore

pandas, `df.info()`, `describe()`

→ Handle Missing Data

Median imputation by category

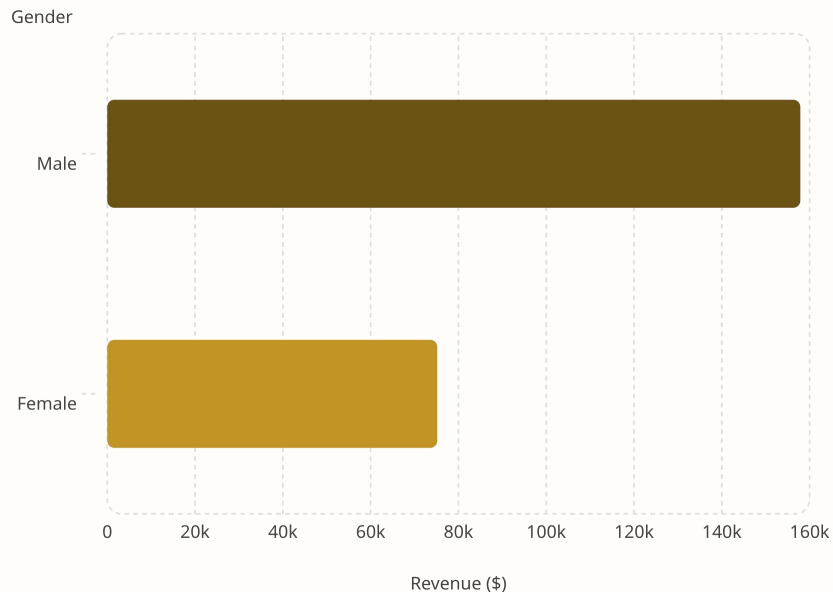
→ Feature Engineering

Created `age_group` &
`purchase_frequency_days`

→ Standardize & Load

Snake case, dropped redundant
columns, loaded to MySQL

Revenue by Gender



Key Insight

Male customers generated **over 2× the revenue** of female customers — a significant gap for targeted marketing.

Product & Shipping Insights



Top-Rated Products

Highest average review ratings identified



Shipping Comparison

Express vs. Standard average purchase amounts



Discount-Dependent Products

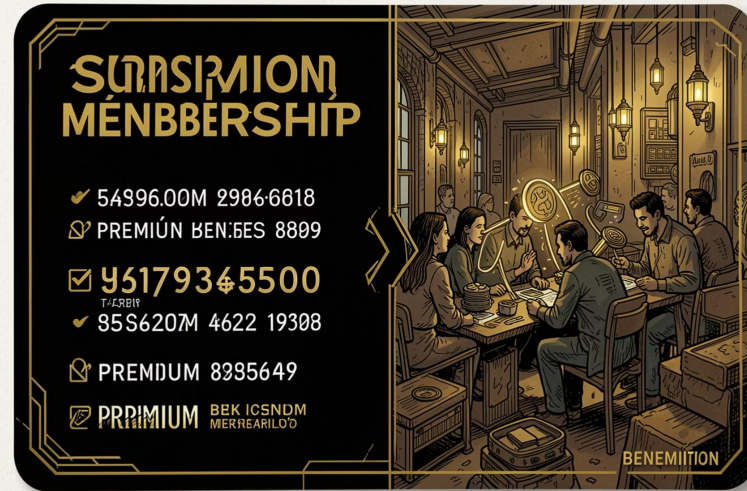
5 products with highest % of discounted purchases

Subscription Behavior

Subscribers vs. Non-Subscribers

Compared **average spend** and **total revenue** across subscription status to reveal purchasing power differences.

- Repeat buyers with >5 **purchases** show higher subscription likelihood.





Customer Segmentation

 New

First-time buyers

 Returning

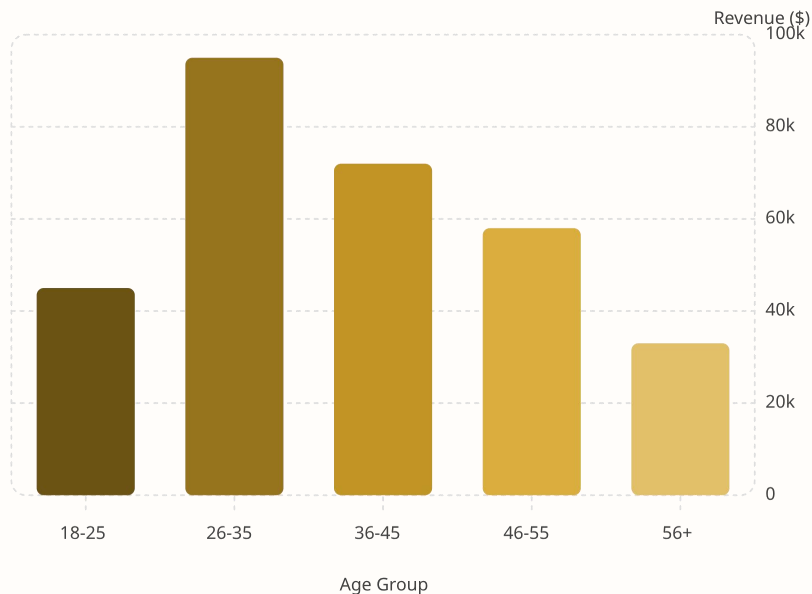
Repeat purchasers

★ Loyal

High-frequency buyers

Segments classified based on purchase history to enable personalized engagement strategies.

Revenue by Age Group



High-Revenue Age Groups

26–35 and 36–45 cohorts drive the majority of revenue — prime targets for campaigns and product positioning.

Power BI Dashboard

Interactive dashboard consolidating all SQL insights — revenue, segments, products, and subscriptions — into a single view for stakeholders.

